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SE

Date / /

Assignment - 01

Q. 1.

Ans

a.) Evolving Role of Software & Software Characteristics

The role of software has evolved from simple data processing to intelligent, autonomous decisionmaking systems. In an AI-based smart surveillance system, software not records video but also detects faces, recognizes behavior, and predicts threats.

Software characteristics influencing design and development

- Reliability: The system must work continuously without failure.
- Scalability: Must handle large video streams from multiple cameras.
- Maintainability: AI models need frequent updates.
- Security: Protection against data breaches.
- Real-time performance: Immediate response to suspicious activity.

b.) Software Myths and Their Negative impact

Software myths are false beliefs that can mislead project teams.

Examples of Myths:

- AI systems will work perfectly once deployed.
- More data automatically means 'better results'.

Negative impacts on AI projects

- Underestimation of testing and validation effort
- Ignoring ethical and bias issues.

Q.2 >

↳ The Software crisis refers to problems like increasing complexity, high cost, delays, and low reliability.

Reasons for Software crisis in AIMI

- Complex algorithms and models
- Large and dynamic datasets
- Continuous learning and updates.

"No Silver Bullet" Justification;

There is no ~~silver~~ single technology or tool that can completely solve AI-based software problems because:

- Problems vary across domains.
- AI Systems involves data, algorithms and ethics.

Q.3 >

↳ Q.1) Software Engineering Phases

1. Requirement Analysis: Identify medical needs, accuracy.
2. Design: System architecture, model selection, data flow.
3. Implementation: Develop AI models and software modules.

4. Testing: Validate accuracy, reliability.

~~5. Develop~~

5. Deployment: Integrate into healthcare environment

6. Maintenance: Update models with new medical data

b.) Team Software Process (TSP)

TSP emphasizes team collaboration and accountability.

Benefits:

- Better coordination between developers, data scientists, and testers.
- Early detection of defects.
- Improved productivity and quality
- Reduced risk in safety-critical system.

Q 4.

<u>Model</u>	<u>Risk Handling</u>	<u>Iteration</u>	<u>Suitability for AI/ML</u>
Waterfall	Poor	No	Not suitable
Prototype	Medium	Limited	Useful for requirement clarity
Spiral	Excellent	Yes	Highly suitable
Iterative	Good	Yes	Suitable

Conclusion

The Spiral Model is most suitable for AI/ML System.

Q5.

4) a.) Why Agile is More Suitable

Agile methodology is better than traditional models because:

- User preferences change frequently.
- Continuous feedback improves recommendation accuracy.
- Supports frequent model updates.
- Faster delivery and flexibility.

b.) Scrum Framework Roles in AI/ML

- Product Owner: Defines business goals and prioritizes features.
- Scrum Master: Ensures smooth scrum process and removes obstacles.
- Scrum Team: Develops models, trains data.

c.) Scrum Sprint vs Kanban & Recommendation

<u>Aspect</u>	<u>Scrum</u>	<u>Kanban</u>
Time-boxed	Yes	No
Continuous flow	No	Yes
flexibility	Medium	High.

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